

DAY 2 · SESSION 4

Resilience, Disaster Recovery, and Multi-Region Design

Azure Storage, Data Protection, and Resilience at Scale

University of Minnesota

This afternoon, block by block

01 **Block 1 · 55 min**

Redundancy options, object replication, and cross-region copy

02 **Block 2 · 50 min**

Azure Site Recovery and the regional loss demonstration

03 **Break · 15 min**

Coffee and a reset before the design method

04 **Block 3 · 40 min**

RPO and RTO design method and Resiliency in Azure

05 **Block 4 · 80 min**

Design exercise and facilitated review

Enterprise data must stay up; research data must never be lost

- Enterprise data must be highly available or the university cannot function
- Research data may not need the same availability, but it must never be lost
- Multiple copies of petabytes are expensive, so every resilience choice needs an honest cost line



By the end of this session you can

1. Explain what LRS, ZRS, GRS, GZRS, RA-GRS, and RA-GZRS guarantee, and what they do not
2. Design object replication and cross-region copy for research data
3. Plan VM disaster recovery with Azure Site Recovery, including test failover and recovery plans
4. Define RPO and RTO per workload class and map them to Azure capabilities
5. Address data residency and copies held outside Azure

Availability, durability, and recoverability are different promises



Availability

Can I read and write right now?



Durability

Will the bytes survive hardware and site failures?



Recoverability

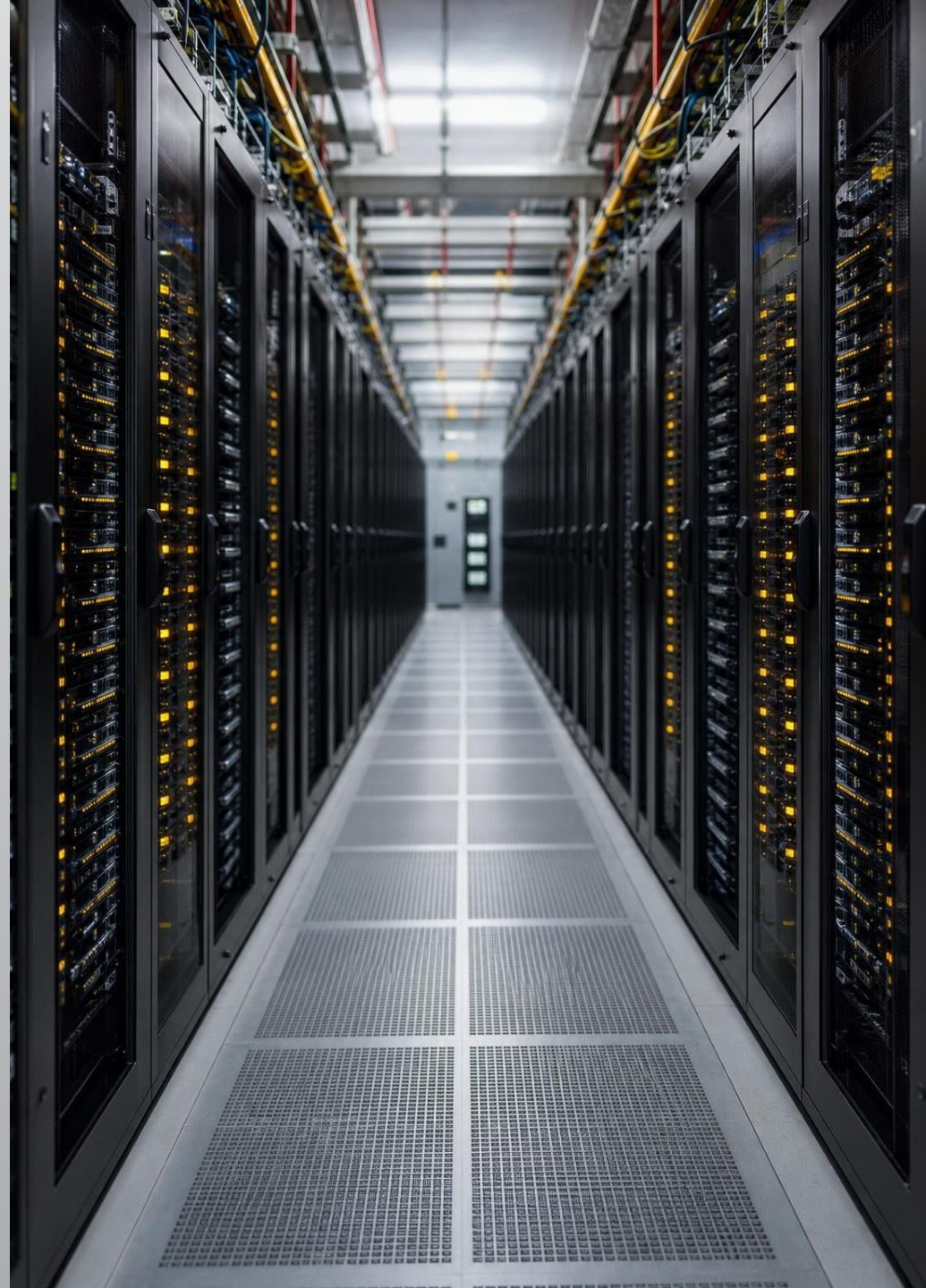
Can I get back to a good state after deletion, corruption, or ransomware?

Redundancy delivers the first two; only backup and immutability (Session 3) deliver the third.

BLOCK 1 · 55 MIN

Redundancy options, object replication, and cross-region copy

What each copy protects, and what it cannot.



Redundancy in the primary region: always three copies

Data is always replicated three times in the primary region.

LRS

Three copies, one datacenter

- At least 11 nines durability over a year
- Lowest cost
- No protection against a datacenter-level disaster such as fire or flood

ZRS

Three zones, synchronous

- Synchronous copies across three availability zones
- At least 12 nines durability
- Stays readable and writable if one zone is lost

Geo-redundancy: a second region, asynchronously

GRS

LRS in the primary plus asynchronous copy to a secondary region

GZRS

ZRS in the primary plus asynchronous copy to a secondary region

RA-GRS / RA-GZRS

Add read access to the secondary during a primary outage

⚠ The secondary is a paired region for most accounts. Replication is asynchronous, so recent writes can be lost on unplanned failover.

What redundancy does not do

It copies mistakes too

Deletes, overwrites, and ransomware encryption replicate to every copy.

It is account-wide

All services in an account share one setting. Isolate data with different redundancy needs in separate accounts.

Pair redundancy with soft delete, versioning, backup, and immutability.

Redundancy options at a glance

Option	Copies and placement	Survives	Read from secondary
LRS	3 in one datacenter	Drive and rack failure	No
ZRS	3 across zones in one region	Zone loss	No
GRS	LRS plus async copy in second region	Regional loss	No
GZRS	ZRS plus async copy in second region	Zone and regional loss	No
RA-GRS	GRS with read access	Regional loss	Yes
RA-GZRS	GZRS with read access	Zone and regional loss	Yes

Account failover: planned and unplanned

PLANNED

Customer-managed planned failover

For DR testing and proactive response. Primary and secondary swap and the account stays geo-redundant.

UNPLANNED

Customer-managed unplanned failover

For an unexpected outage. Data not yet replicated can be lost.

- Check the last sync time to estimate potential data loss before failing over
- Plan failback and understand features that conflict with failover

Object replication source and destination accounts cannot use customer-managed failover, NFS 3.0 accounts support only unplanned failover, and immutability policy changes made after the last sync time do not carry over.

Object replication: selective, asynchronous copies of block blobs



What it does

Asynchronously copies block blobs from a source account to up to two destination accounts in any region, subscription, or tenant. Cross-tenant replication is disabled by default for accounts created after December 15, 2023.



Prerequisites

Change feed on the source and blob versioning on both accounts; GPv2 or premium block blob accounts; up to 1,000 rules per policy.



Not supported

Hierarchical namespace accounts, NFS 3.0 or SFTP accounts, append and page blobs, snapshots, blobs already in Archive on either side, and accounts that use customer-managed failover.



Lag and cost

Asynchronous by design; priority replication adds an SLA of 99 percent of objects within 15 minutes on the same continent. Replicate while online, then use a lifecycle policy to tier the destination copy to Archive.

Object replication versus geo-redundancy

	Geo-redundancy	Object replication
Scope	Whole account	Chosen containers and prefixes
Destination	Fixed secondary	Any destination region, in its own tier and redundancy
Management	Microsoft-managed	A separately administered second account

- Object replication is useful for "not all eggs in one basket"
- Replication is asynchronous; plan for lag, verify it with replication status, and enable priority replication where a 15-minute objective matters.



Copies beyond Azure: second cloud or on premises

- AzCopy or Azure Data Factory can copy data to an on-premises target or another provider on a schedule
- Gives a copy that is logically and administratively separate from Azure
- Budget for egress on every copy out of Azure
- Keep integrity manifests (Session 2) so remote copies can be verified

BLOCK 1 CHECKPOINT

Sort your data classes

01

Which data classes need to survive a regional loss?

02

Which need to stay available during one?

03

Which need a copy outside Azure entirely?

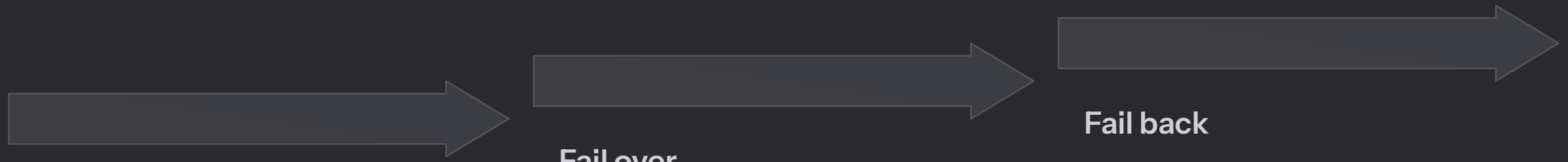
BLOCK 2 · 50 MIN

Azure Site Recovery and the regional loss demonstration

From replication to a rehearsed failover.



Azure Site Recovery: replicate, fail over, fail back



Replicate

Azure VMs between regions and on-premises machines to Azure

Fail over

During an outage

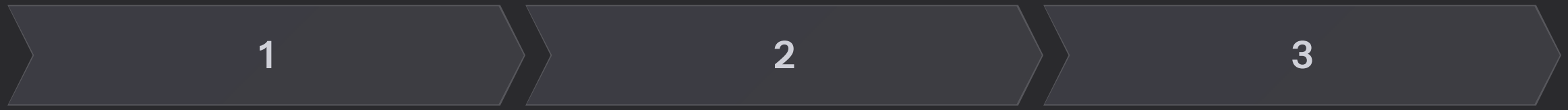
Fail back

When the primary returns

- Replication can now be set between any two Azure regions, not only within a continent
- Replication data is held in a Recovery Services vault

Recovery plans: model the application, not the server

Recovery groups define failover order and startup sequence, up to 100 protected instances per plan.



Database tier

Starts first.

Middleware

Starts once data is up.

Web front end

Opens to users last.

Add manual instructions and scripted tasks to automate recovery and reduce RTO.

Test failover: rehearse without risk

Safe by design

Validates replication and DR strategy without data loss or downtime.

Does not affect ongoing replication or production.

How to run it

Run on a single VM or a whole recovery plan.

Use an isolated test network, then clean up the test failover.

Choosing the recovery point trades RPO against RTO

Latest processed

Lowest RTO, no time spent processing data.

Latest app-consistent

Latest application-consistent point.

Latest

Processes all sent data first for the lowest RPO.

Latest multi-VM processed

For plans with multi-VM consistency.

Network mapping keeps failed-over VMs connected

- Maps source virtual networks to target networks in the recovery region
- Create mappings in both directions for failover and failback
- If not prepared, Site Recovery creates a target network with an -asr suffix
- Plan IP addressing, DNS, and private endpoints in the target region

Demonstration: regional loss, simulated and survived

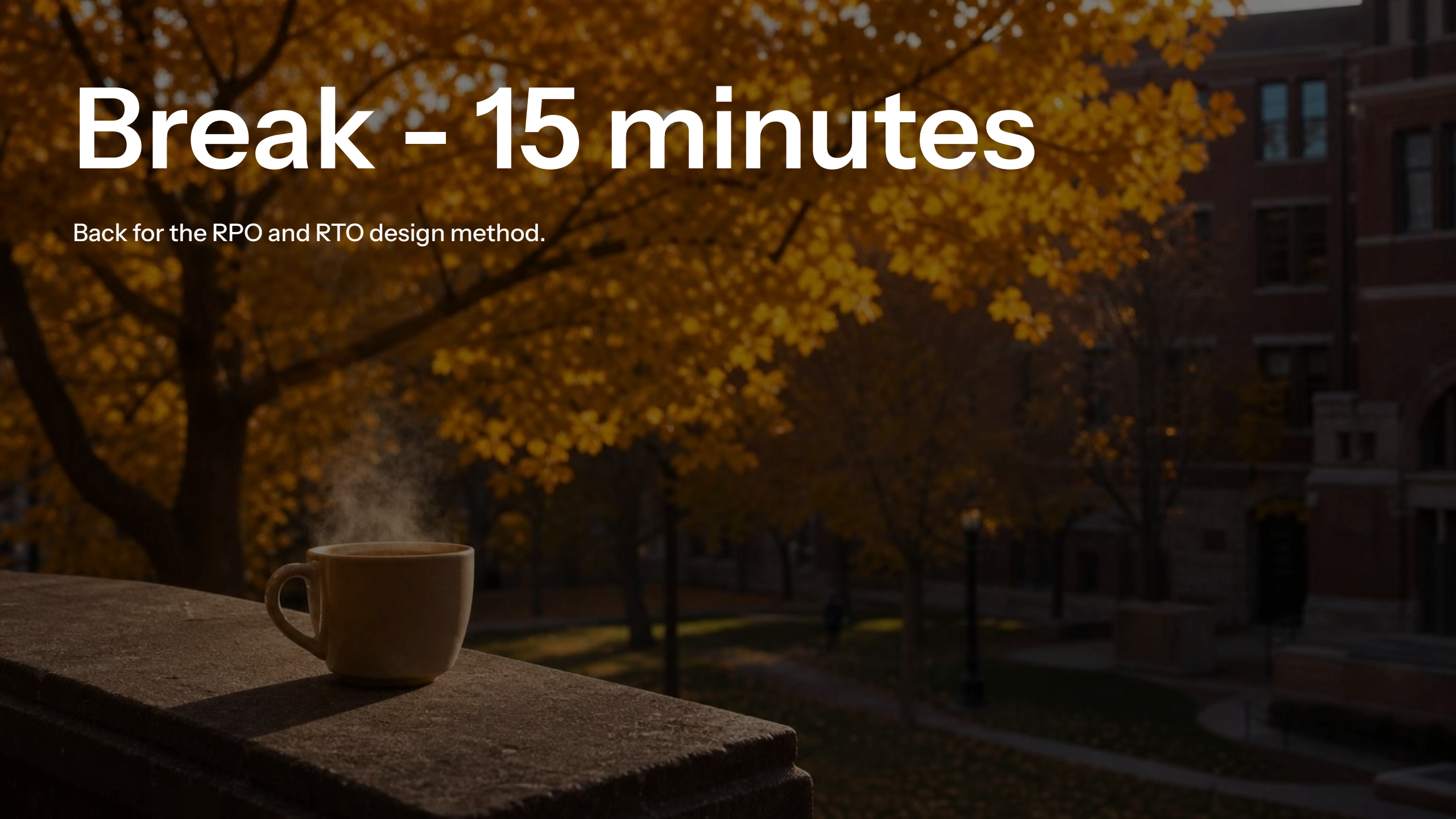
1. Initiate account failover for a GZRS storage account
2. Run a test failover of a protected VM through a recovery plan
3. Show the resilience dashboard before and after
4. Compare the monthly cost delta between the resilience tiers used

A region disappears and the workload returns in another, with every protected item still compliant.



Break – 15 minutes

Back for the RPO and RTO design method.



BLOCK 3 · 40 MIN

RPO and RTO design method and Resiliency in Azure

Turning business tolerance into platform choices.



RPO and RTO in plain terms

RPO

How much data you can afford to lose, measured in time

RTO

How long the workload can be down

- Both are business decisions; the platform choice follows from them
- Tighter targets cost more in copies, compute, and operational effort

The design method in five steps

01

Classify

Workloads by business impact

02

Target

Tiered RPO and RTO per class

03

Map

Each tier to redundancy, replication, backup, and DR

04

Price

Each tier

05

Document

The accepted risk and who accepted it

Mapping targets to Azure capabilities

Tier	Typical need	Azure building blocks
Mission critical	Minimal data loss, fast recovery	ZRS or GZRS, Site Recovery, immutable backup
Business important	Hours of tolerance	GRS, backup with cross-region restore
Research active	Must not lose data, can wait	Flat accounts: GRS or object replication, versioning, soft delete. HNS data lakes: GRS plus vaulted backup for Data Lake Storage
Research archive	Never lost, days to recover	Container-level WORM, a second copy made while the data is online (object replication for flat accounts; AzCopy or Data Factory for HNS accounts or data already in Archive), then Archive tier at both ends; a copy outside Azure for independence

Resiliency in Azure: one view of protection

Azure Business Continuity Center is now Resiliency in Azure: zone resiliency, high availability, backup and DR, and ransomware protection in one platform.



Unified visibility across backup and Site Recovery replication



Protection status reporting and detection of unprotected resources



Prove compliance with the tier assigned to each workload



Data residency and independence

- Choose regions with residency, sovereignty, and contractual obligations in mind
- Confirm the secondary region of geo-redundant storage meets the same obligations
- Keep a copy logically and administratively separate from the primary environment
- Separate subscriptions, tenants, or providers limit a single point of compromise

BLOCK 4 · 80 MIN

Design exercise and facilitated review

Six workloads, one matrix, defended in the room.



Design exercise: the university resilience matrix



Enterprise database



Identity platform



Learning management data



Active research dataset



Cold research archive



Departmental file share

For each workload select redundancy, replication, backup, and DR approach; assign RPO and RTO; estimate relative cost.

How the 80 minutes run

01

Form groups

Take the resilience matrix worksheet

02

Complete

All six workloads

03

Defend

Groups defend their choices

04

Review

Facilitated review against the Well-Architected Framework reliability pillar

Review lenses

Does the RPO and RTO match the business impact?

Does redundancy cover the failure you actually fear?

Is there a recoverable copy outside the blast radius?

Is the cost line honest, and is the accepted risk written down?

Key takeaways and what comes next

- Redundancy protects against infrastructure failure, not deletion or corruption
- Object replication and copies outside Azure add independence at a known cost
- Site Recovery plans and test failovers turn DR from hope into rehearsal

MATERIALS

Demo script, resilience matrix worksheet, redundancy comparison card, sample recovery plan

NEXT, DAY 3

Next-generation protection for Oracle and enterprise databases

Capstone input: University resilience matrix applied to six workload classes